8. Write a Python program to model a segmented network's access policy and demonstrate how network segmentation blocks lateral movement from a compromised low-trust segment into a sensitive one.

```
[10]
✓ 24s

allowed = {
    ("Low-Trust", "Application"),
    ("Authorized Admin", "Sensitive")
}

segments = [
    "Low-Trust",
    "Application",
    "Sensitive",
    "Authorized Admin",
    "Unknown"
]

print("Network Segmentation Test")
print("Available segments:", ", ".join(segments))

source = input("Enter source segment: ").strip()
destination = input("Enter destination segment: ").strip()

if source not in segments:
    print("Access Denied: Unknown source segment")

elif destination not in segments:
    print("Access Denied: Unknown destination segment")
```

```
[10]
✓ 24s

source = input("Enter source segment: ").strip()
destination = input("Enter destination segment: ").strip()

if source not in segments:
    print("Access Denied: Unknown source segment")

elif destination not in segments:
    print("Access Denied: Unknown destination segment")

elif (source, destination) in allowed:
    print(source, "->", destination, ": Access Allowed")

else:
    print(source, "->", destination, ": Access Denied")

Network Segmentation Test
Available segments: Low-Trust, Application, Sensitive, Authorized Admin, Unknown
Enter source segment: Low-Trust
Enter destination segment: Application
Low-Trust -> Application : Access Allowed
```